Samson Wu

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EDUCATION

Northeastern University — Boston, MA
Master of Science in Robotics (Khoury College of Computer Sciences) — Sep 2026 – Present
Related courses: Reinforcement Learning, Robotic Sensing and Navigation — Expected Graduation: Dec 2028
Focus: Mobile/Loco-manipulation and dexterous manipulation of robots

University of Toronto — Toronto, ON
BASc in Engineering Science, Major in Machine Intelligence — Sep 2021 – May 2026
Honours: Dean's List (7 semesters) — GPA (Last 2 years): 3.78
Related courses: Machine Learning, Control and Robotics, Computer vision

TECHNICAL KNOWLEDGE

Languages: Python, Java, SQL, C
Software: Mujoco (Robocasa), Pytorch, Tensorflow, ROS2, GCP, AWS, PyCharm, DB Visualizer, SSMS, Apache Airflow, Jira, Postman
Other Skills: JAX, Fusion360, MATLAB, PySpark, Springboot, React, T-SQL, Devops and CI/CD

PROJECTS

Autonomous robot for unloading and loading a dishwasher, Thesis — Sep 2025 – May 2026

- Conducting a comprehensive survey on mobile manipulation frameworks, spatial representations, and robot learning paradigms to establish baseline architecture.
- Evaluated, and trained multiple sample-efficient Point-cloud models in Pytorch to predict rack occupancy for dishwasher loading and unloading, achieving 80%+ accuracy on custom synthetic dataset of 1200+ RGBD.
- Applied research to integrate all systems (localization, rack modelling, CNN model, Vision Foundation models, grasp candidates) in an autonomous pipeline to achieve 70% load success rate in simulation.

Generating Animal Hybrids Using Adversarial Autoencoders, Project — Feb 2024 - Apr 2024

- Collaborated in a team of three to research and write a literature review on strategies for low-level image representation and interpolation of images using autoencoders and adversarial networks.
- Helped to develop, train, and tune GAN-like AEAI and ACAI networks on dataset of 10000+ images of animal faces with cloud compute. Demonstrated smooth interpolation of animal features of pose in some cases.

EXPERIENCE

AI Developer @ Geotab, Toronto, ON — May 2026 – Aug 2026

- Optimized workflow, improved UX, and increased determinism for internal AI Agent SDLC workflow (Claude Code plugin), overall increased development speed by 5-10 times compared to humans.
- Engaged in rapid development and support for a multi-cli internal AI harness (runtime) tool used by the entire organization daily for enhanced productivity. (~2000 monthly active users, 60% of organization)
- Designed and implemented a pipeline for automatic skill demotion using GCP and Gitlab infrastructure, eliminating hours of weekly manual labor.

Data Platform Developer @ Geotab, Toronto, ON — May 2024 – Apr 2025

- Engineered a distributed Ray pipeline to parallelize heavy video frame ingestion, validating scalable asynchronous workload processing for high-throughput video pipelines.
- Developed, fixed bugs, vulnerabilities, and wrote unit tests for a microservices-based Java Spring Boot video processing cloud application in GCP to add features and increase code coverage.
- Designed and implemented automated monitoring cronjobs deployed in the cloud using SQL and Python to measure performance of the application and provide alerting in production.

Software Engineer Co-op @ Sunlife, Toronto, ON May 2023 – August 2023

- Designed and developed UI for an internal database management portal using React libraries, Javascript, HTML, and CSS in an agile environment.
- Developed backend API endpoints in Java to enable functionality of pages, using GET, PUT, and POST requests to interact with the SQL DB. Released portal for internal use by end of term on UAT.
- Debugged, tested, and created stored procedures using T-SQL in SSMS for robust database management.